

Sealed-Typen & Pattern Matching

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Was nervt Sie an diesem Code?

```
interface Expr {}  
record IntLiteral(int value) implements Expr {}  
record Add(Expr left, Expr right) implements Expr {}  
record Negate(Expr inner) implements Expr {}  
  
int evalOld(Expr e) {  
    if (e instanceof IntLiteral) {  
        IntLiteral lit = (IntLiteral) e;  
        return lit.value();  
    } else if (e instanceof Add) {  
        Add add = (Add) e;  
        return evalOld(add.left()) + evalOld(add.right());  
    } else if (e instanceof Negate) {  
        Negate neg = (Negate) e;  
        return -evalOld(neg.inner());  
    } else {  
        throw new IllegalArgumentException("Unknown expression: " + e);  
    }  
}
```

Pattern Matching für `instanceof`

```
// vor Java 16:  
if (obj instanceof String) {  
    String s = (String) obj;  
    if (s.length() > 5) {  
        IO.println(s.toUpperCase());  
    }  
}
```

```
// modern:  
if (obj instanceof String s && s.length() > 5) {  
    IO.println(s.toUpperCase());  
}
```

Switch Expressions

```
switch (day) {  
    case MONDAY:  
    case FRIDAY:  
        IO.println("Almost weekend");  
        break;  
    default:  
        IO.println("Another day");  
}
```

```
String message = switch (day) {  
    case MONDAY, FRIDAY -> "Almost weekend";  
    case SATURDAY, SUNDAY -> "Weekend!";  
    default -> "Another day";  
};
```

Type Patterns in `switch`

```
sealed interface Expr permits IntLiteral, Add, Negate {}
```

```
record IntLiteral(int value) implements Expr {}
```

```
record Add(Expr left, Expr right) implements Expr {}
```

```
record Negate(Expr inner) implements Expr {}
```

```
int eval(Expr e) {  
    return switch (e) {  
        case IntLiteral lit -> lit.value();  
        case Add add -> eval(add.left()) + eval(add.right());  
        case Negate neg -> -eval(neg.inner());  
    };  
}
```

Guarded Patterns im `switch`

```
int sign(Expr e) {  
    return switch (e) {  
        case Expr.IntLiteral lit when lit.value() > 0 -> 1;  
        case Expr.IntLiteral lit when lit.value() < 0 -> -1;  
        case Expr.IntLiteral lit -> 0;  
        default -> throw new IllegalArgumentException("Not a literal: " + e);  
    };  
}
```

Record-Patterns / Dekonstruktion

```
record Point(int x, int y) {}
```

```
static int manhattan(Object o) {  
    if (o instanceof Point) {  
        Point p = (Point) o;  
        return Math.abs(p.x()) + Math.abs(p.y());  
    }  
    throw new IllegalArgumentException("Not a point: " + o);  
}
```

Record-Patterns / Dekonstruktion

```
record Point(int x, int y) {}
```

```
static int manhattan(Object o) {  
    if (o instanceof Point) {  
        Point p = (Point) o;  
        return Math.abs(p.x()) + Math.abs(p.y());  
    }  
    throw new IllegalArgumentException("Not a point: " + o);  
}
```

```
static int manhattan(Object o) {  
    return switch (o) {  
        case Point(int x, int y) -> Math.abs(x) + Math.abs(y);  
        default -> throw new IllegalArgumentException("Not a point: " + o);  
    };  
}
```


Für Datenhierarchien (z.B. `Expr`, `Shape`, `Command`) nutzen Sie heute:

- `record` für Daten,
- `sealed` für eine abgeschlossene Variantenmenge, und
- Pattern Matching im modernen `switch` (mit Arrow-Syntax) für knappen, typsicheren und gut wartbaren Code im (teilweise) funktionalen Stil.



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